



Smart Home Security and Privacy Mitigations: Consumer Perceptions, Practices, and Challenges

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Abstract. As smart home technology is becoming pervasive, smart home devices are increasingly being used by non-technical users who may have little understanding of the technology or how to properly mitigate privacy and security risks. To better inform security and privacy mitigation guidance for smart home devices, we interviewed 40 smart home users to discover their security and privacy concerns and mitigation strategies. Results indicated a number of concerns, but a general willingness to accept risk in lieu of perceived benefit. Concern was sometimes, but not always, accompanied by users taking mitigating actions, although most of these were simplistic and not technical in nature due to limited options or lack of user technical knowledge. Our results inform how manufacturers might empower users to take protective actions, including providing security tips and more options for controlling data being collected by devices. We also identify areas that might benefit from third-party involvement, for example by providing guidance to manufacturers on minimum privacy and security standards or developing a security and privacy rating system to aid users in selecting devices.

Keywords: Smart home · Internet of Things · Security · Privacy · Usability

1 Introduction

As Internet of Things (IoT) smart home technology is becoming pervasive, smart home devices are increasingly being used by non-technical users [10] who may have little understanding of the technology or awareness of the implications of use, including considerations for privacy and security. Since their inception, smart home devices have become the target of security attacks, placing consumers' data, privacy, and safety at risk [13, 16]. In addition, concerns about the privacy and protection of potentially sensitive consumer data are surfacing [6, 12]. In fact, the U.S. Federal Bureau of Investigation (FBI) recently issued security and privacy warnings about smart televisions and other IoT devices [18, 19]. Therefore, it is critical

that users are provided with the means to safeguard their information and households while still enjoying the convenience of these devices.

Unfortunately, smart home device manufacturers may not provide privacy and security protections and configuration options [4], or, if they do, these options may not be transparent to the user. In addition, smart home users may not be knowledgeable enough to discern which mitigations would be most effective, or may only implement simplistic mitigations that might be inadequate [1, 11, 15, 26]. This inadequacy was demonstrated by recent stories of weak user-configured passwords being responsible for parents and children being surveilled and terrorized after their smart home devices were exploited [13].

Understanding consumers' interactions with smart home devices and their current privacy and security mitigation strategies is a first step towards developing guidance for manufacturers and third-party organizations to aid consumers. We sought to gain this understanding via an in-depth interview study of 40 smart home consumers to discover their overall experiences with, perceptions of, and challenges regarding their smart home devices. This paper addresses a subset of research questions (RQs) from the broader study that were focused on security and privacy:

RQ1: What are smart home users' privacy and security concerns, if any?

RQ2: What mitigation actions, if any, do users take to address their concerns?

RQ3: What are the factors affecting users' implementation (or lack of implementation) of privacy and security mitigations?

RQ4: What do users want (actions to take on their own or from others) in order to feel like their privacy and security are adequately protected?

We found that many users have privacy and security concerns but are mostly implementing simplistic mitigations to counter those concerns. However, some smart home users displayed a lack of concern or failed to take mitigation actions even if they do have concerns. The interviews revealed several challenges to the implementation of effective security mitigations, including users having incomplete threat models, privacy resignation, lack of transparency, poor usability of privacy and security-related device features, and lack of user technical knowledge to discern or implement appropriate mitigations. Our study makes several contributions:

- We confirm and expand upon prior studies that investigated smart home users' privacy and security concerns and mitigations [3, 20, 26] with a larger, more diverse participant sample.
- We identify several mitigations not previously described in the literature, including a more in-depth examination of smart home device updates.
- We distill participants' privacy and security "wishlist," which provides insight into potential areas for improvement in smart home device design and data handling.
- Our results inform how manufacturers and third-party evaluators might provide a more usable security and privacy experience.

2 Related Work

Prior work has examined perceptions of smart home privacy and security. Security and privacy concerns can be barriers to adoption of smart home devices. Lau et al. find that some non-users are privacy conscious and distrustful of privacy and security of smart home devices and their manufacturers, and that smart home devices generally cross these non-users' perceived privacy thresholds [11]. This finding is corroborated by Parks Associates [14], Worthy et al. [25], Emami-Naeini et al. [3], and Fruchter and Liccardy [7], who find that a lack of trust in vendors to properly safeguard personal data is a major obstacle to adoption of smart home technology. From a broader IoT perspective, Williams et al. [24] found that IoT is viewed as less privacy-respecting than non-IoT devices such as desktops, laptops, and tablets.

Adopters were found to share the same concerns, and often expressed a lack of agency in the control of their data [11]. However, they generally have higher tolerances for privacy violations, and willingly or reluctantly accept the trade-off in exchange for the convenience and utility offered by smart home devices [11]. They are generally more trusting towards well-known manufacturers and often express that they have “nothing to hide” [11, 20]. They also have complex, but incomplete threat models, which includes a general sense of being surveilled by manufacturers or the government, and the possibility of being attacked by hackers, but a lack of awareness of botnets and the sale of inferred data [1, 3, 27]. A main security concern was the possibility of a breach in the cloud that would expose user data [20].

Multiple studies discovered both technical and non-technical mitigations to address security and privacy concerns, for example passwords, secure configurations for the home network, and altering behavior around the devices [1, 11, 15, 20, 26]. However, they also identified lack of action. Reasons may be lack of awareness and availability of these options, privacy resignation, trust in the manufacturers, and assignment of responsibility to entities other than the users themselves [11, 20, 26].

Our study confirms many of the findings identified in prior literature while identifying additional mitigations such as device selection, access control, and updates. In addition, unlike other studies, we collected a wish list of mitigations that can help inform manufacturers and other entities in making privacy and security protections for smart home devices more usable for consumers.

3 Methods

We conducted semi-structured interviews of 40 smart home consumers to understand their perceptions of and experiences with smart home devices from purchase decision, to implementation, to everyday usage. The in-depth interviews afforded more detailed data than could be collected via anonymous surveys and the ability to ask follow-up questions to explore responses [2]. To protect participants' confidentiality, data were recorded with generic identifiers (such as P10)

and not linked back to individuals. The study was approved by the National Institute of Standards and Technology (NIST) research protections office.

We hired a consumer research company to recruit 33 general public participants, and identified seven participants via professional contacts. To determine study eligibility, adult participants interested in the study completed an online screening survey about their smart home devices, role with the devices (i.e., decision maker, purchaser, installer, administrator, troubleshooter, or user), professional background, basic demographic information, and number of household members. To ensure information-rich cases, we then purposefully selected participants who had two or more smart home devices for which they were active users.

The interview protocol addressed the following areas: understanding smart home terminology, purchase and general use, likes and dislikes, installation and troubleshooting, privacy, security, and physical safety. Interviews lasted an average of 41 min. Prior to the interviews, we informed the participants about the study and how we would protect their data by not recording any personal identifiers that could be linked back to the participant. All interviews were audio recorded and transcribed. General public participants were compensated with a \$75 gift card.

Using widely-accepted qualitative data analysis methods [9], all three authors individually coded a subset of four interviews, then met to develop and operationalize a codebook to identify concepts within the data. Based on the codebook, we then performed iterative coding on the remainder of the interviews, with two coders per transcript. Each pair of coders met to discuss and resolve areas of difference in code application. As a group, we then progressed to the recognition of relationships among the codes and examined patterns and categories to identify themes. In this paper, we focus on themes related to privacy and security mitigations and concerns.

4 Participant Demographics

We interviewed 40 participants, 32 of whom were the installers and administrators of the devices (indicated with an A after the participant ID) and eight who were non-administrative users of the devices (indicated with a U). 55% were male, and 45% were female. Multiple age ranges were represented, with the majority (70%) between the ages of 30 and 49. Overall, participants were highly educated with all but one having at least a bachelor's degree and almost half (45%) having at a graduate degree. Table 1 shows participant demographics.

All but one participant had three or more individual smart home devices, with 34 (85%) having three or more different types of devices. Figure 1 shows the general categories of smart home devices in participants' homes. Represented categories, along with examples of devices in that category, were:

Table 1. Participant Demographics. ID: A - smart home administrators/installers, U - smart home users; Education: M - Master's degree, B - Bachelor's degree, C - some college, H - High school.

ID	Gender	Age	Education	Occupation
P1_A	F	50–59	M	Liaison
P2_A	M	30–39	M	Lead engineer
P3_A	F	40–49	M	Professor
P4_A	M	60+	M	Retired
P6_U	F	30–39	B	Events manager
P7_A	M	30–39	B	Software engineer
P8_A	M	30–39	B	Federal employee
P9_A	F	30–39	M	Educationist
P10_A	M	30–39	B	Computer scientist
P11_A	M	50–59	M	Electrical engineer
P12_U	F	30–39	M	Administrative assistant
P13_A	M	50–59	M	Manager, Cognitive scientist
P14_U	F	40–49	H	Information specialist
P15_A	M	30–39	B	Computer scientist
P16_A	M	40–49	M	Research chief
P17_A	F	30–39	M	Systems engineer
P18_A	M	30–39	B	Business consultant
P19_A	M	50–59	B	Retail services specialist
P20_A	F	30–39	B	Administrator
P21_U	F	18–29	B	Human resources manager
P22_A	M	30–39	B	Executive admin assistant
P23_A	F	40–49	M	Community arts specialist
P24_A	M	40–49	B	Operational safety analyst
P25_A	M	30–39	B	Program management analyst
P26_A	M	30–39	B	Analyst
P27_A	F	40–49	M	Program coordinator
P28_A	F	50–59	B	Consultant
P29_A	M	18–29	M	Events coordinator
P30_U	F	18–29	B	Event planner
P31_A	F	30–39	M	Lobbyist
P32_A	M	30–39	B	Health educator
P33_A	M	18–29	B	Senior technology analyst
P34_A	M	40–49	B	Financial analyst
P35_A	M	40–49	M	Accountant
P36_A	F	30–39	B	Project manager
P37_A	F	40–49	M	Assistant principal
P38_U	F	60+	M	Special educator
P39_U	M	60+	M	Retired
P40_U	F	30–39	C	Customer service rep
P41_A	M	40–49	B	Security

Smart security: security cameras, motion detectors, door locks.

Smart entertainment: smart televisions, speakers, streaming devices, other connected media systems.

Home environment: smart plugs, energy monitors, lighting, smoke and air quality sensors, thermostats.

Smart appliances: refrigerators, coffee pots, robot vacuums, washers.

Virtual assistants: voice-controlled devices such as Amazon Echo (colloquially called Amazon Alexa) and Google Home.

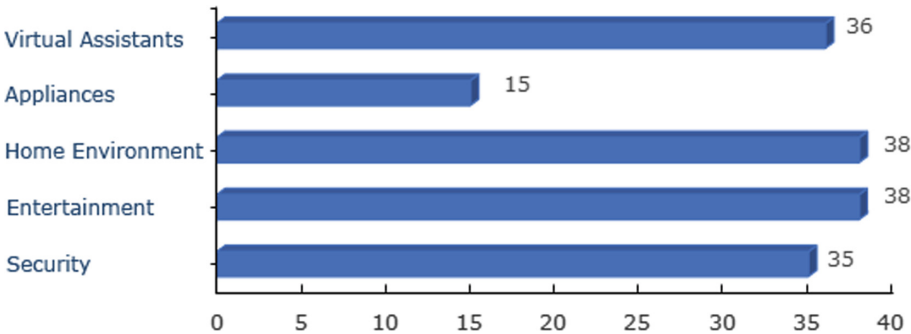


Fig. 1. Types of Smart Home devices owned by participants.

5 Results

In this section, we report results from a subset of the interview data specific to privacy and security concerns, mitigations, and mitigation wish lists. Counts of the number of participants mentioning various concepts are provided in some cases to illustrate weight or unique cases and are not an attempt to reduce our qualitative data to quantitative measures.

5.1 Concerns

We present an overview of concerns identified in our study to provide context for what our participants believe might need to be addressed by mitigations. Participants' privacy and security concerns are summarized in Table 2. For each concern in the table, we include whether the concern was discussed in a privacy or security context (or both), the number of participants mentioning each concern, and an example participant quote to illustrate the concern.

The most frequently mentioned concerns that were discussed within both the privacy and security contexts included: audio and video access via smart home devices such as virtual assistants and cameras; data breaches of the manufacturer; foreign and domestic government access to data; and exposure of financial information via smart home device credentials and apps. Participants

Table 2. Smart Home Privacy and Security Concerns. # - number of participants mentioning the concern

	Concern	#	Example participant quote
Security and Privacy	Audio/video access	34	<i>"I was reading some article where [a virtual assistant] listens in on some of the conversations we have in our house without it being awake... That kind of freaks me out in the sense that we could be talking about something, and they have that information." (P21_U)</i>
	Data breaches	17	<i>"Manufactures can say they can protect things, but in reality, if someone wants something bad enough, I don't know if they really can." (P33_A)</i>
	Government access	12	<i>"I would hate to sound like a conspiracy theorist, but I'm pretty sure the government and places like that can actually see what you do." (P14_U)</i>
	Exposure of financial information	8	<i>"I wouldn't want anybody committing fraud and taking my credit card information to do things they shouldn't be doing." (P37_A)</i>
Privacy	Household profiling	19	<i>"If someone was in control of this [device], they might be able to know what my schedule is, when I'm usually home, when the house is empty." (P34_A)</i>
	Selling data	17	<i>"That's what I'm really afraid of, is them packaging my information to get trends and marketing it." (P13_A)</i>
	Unknowns of data collection	16	<i>"I'm concerned because I think we're unaware of the types of information that these smart devices store of us or have of us." (P21_U)</i>
Security	Device hacking	22	<i>"There's some just people who are really smart and they're sitting somewhere, all they're thinking about is how to get into stuff... And if people could hack into the Department of Defense, they can hack into yours." (P28_A)</i>
	Safety	17	<i>"It could be life threatening... If you rely on the smart device to keep your home locked,... if it does malfunction, there could be extreme circumstances." (P19_U)</i>
	Gaining Wi-Fi access	6	<i>"Many of these devices, you're giving it your network password, so it has full access to everything on your network." (P11_A)</i>
	Linked accounts	4	<i>"If you use a password commonly across different accounts, the same password, if that gets hacked... If I log into my Google account they might be able to get in because I might use the same exact password and user name." (P2_A)</i>
	Poor default security settings	2	<i>"I would be disturbed if I saw a device that, for example, had a password you couldn't change or restricted you to something like a 4-digit key code that's more easily hacked." (P15_A)</i>
	Update issues	2	<i>"I guess one area where I would be worried about would be adding features that may threaten my privacy and security." (P15_A)</i>

talked about the following privacy-specific concerns: household habit profiling; the selling of data and targeted ads; and unknowns about what data is being collected and how it is being used. Security-specific concerns included: general exploitation/hacking of devices; physical security/safety; gaining access to the Wi-Fi network and other devices on that network via smart home devices; gaining access to linked accounts (e.g., email or social media accounts) by exploiting device apps; poor default security settings (e.g., default passwords); and updates potentially having harmful consequences.

We also found examples of various levels of lack of concern, with seven participants having neither privacy nor security concerns. In 24 cases, participants did not think that the information collected by smart home devices was valuable or interesting to others. For example, one participant commented, *“I live a life that you could probably watch. I could probably have cameras in my house, and I wouldn’t feel guilty about that. . . That’s a concern I know some people have. But I didn’t have an issue with that”* (P2_A). We also identified evidence of participants exhibiting privacy and security resignation [11, 17] (8 participants). They are of the opinion that, since so much of their data is already publicly available via other means (e.g., social media, data breaches), smart home devices pose no additional risk. One smart home user said, *“I do dislike having all of my information out there, but I think that, regardless of these smart devices, it’s already out there”* (P17_A). Finally, five participants viewed exploitation of devices (hacking) as a low-probability event. This feeling was often tied to them not valuing information collected by smart home devices: *“Somebody would have to pluck us at random to really be at risk”* (P25_A).

Ultimately, even if they had concerns, participants were more than willing to accept privacy and security risks because of the perceived benefits. One participant commented, *“It’s an acceptable risk if you don’t think you’re doing anything that’s illegal or bad. It’s not like I do anything weird in front of the TV besides exercise, and nobody wants to see that”* (P14_U). Another said, *“It makes my life easier, so I will continue to do it unless I have a major security concern that comes up”* (P17_A).

5.2 Mitigations

Our study discovered a variety of mitigations that participants or others in their household implement to address privacy and security concerns. All mitigations were mentioned in both the privacy and security contexts. Figure 2 shows the number of participants mentioning each mitigation. We describe the mitigations in more detail below.

Authentication. Participants mentioned using various forms of authentication (e.g., passwords, face recognition, two-factor) when asked what actions they take to address their concerns. However, this action was typically not a user choice, but rather prompted during installation. Authentication was most often referenced with regards to the device companion apps, which are often controlled via a cellphone.

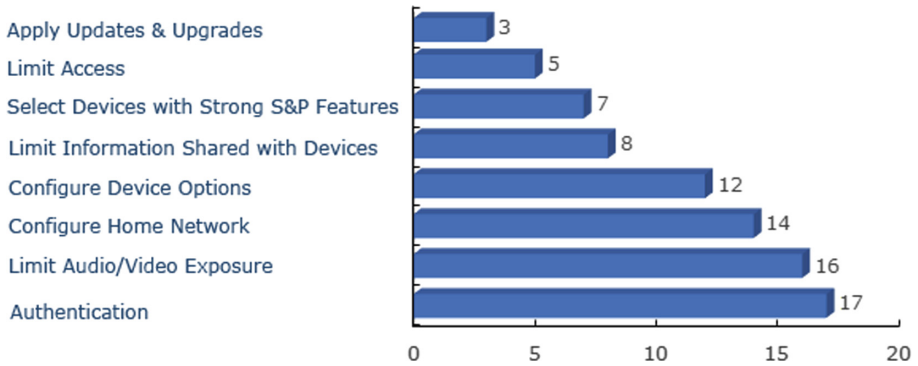


Fig. 2. Security and Privacy mitigations mentioned by participants.

Passwords were the most common authentication mechanism afforded by device companion apps, and often the only mitigation mentioned. One participant said that he addressed his concerns by *“password protecting the devices so nobody can connect to them... It’s not very convenient, but... that’s what I need to do”* (P20_A). Several participants specifically discussed their attempts at having strong passwords: *“I have my own unique passwords that aren’t dictionary words, so that’s how I mitigate”* (P10_A). Another participant used a password manager for her smart home device apps. Two others said that they made sure that they change any default passwords during installation.

Only one participant mentioned two-factor authentication in the context of mitigations: *“If I know that I can do two factor authentication for something, I’ll do that”* (P2_A). When asked about how they authenticate to their devices in a later, separate question, only one additional participant mentioned two-factor authentication, which was an option offered by his smart thermostat.

Limiting Audio and Video Exposure. To address concerns about audio and video being exposed to manufacturers or unauthorized users, study participants mostly mentioned non-technical mitigations. They were careful about where they placed cameras and virtual assistants, avoiding more private rooms in the house. For example, one participant talked about the location of his virtual assistant: *“Bedrooms are just a little more personal. I make sure not to keep it there because... if it does record, I don’t want maybe those conversations and things that happened in the bedroom to be on there”* (P32_A). Several participants were also cognizant of not having sensitive conversations in the vicinity of listening devices: *“I try to keep [my virtual assistant] in a central location and kind of avoid being close to it when having certain conversations”* (P22_A). Others covered cameras not being used. For instance, a participant remarked that her husband took action: *“The [virtual assistant] device has a video camera that you can use, but he’s taped it over”* (P1_A). Finally, several users turned off devices in certain circumstances. One user talked about how her husband

unplugs their virtual assistants when he is teleworking to guard against potentially sensitive conversations being recorded. Another said, *“With the security camera, sometimes I switch it off... It’s when I’m really like out of town, that’s when I like to switch it on back again”* (P34_A).

Network Configuration. The security and privacy of smart home devices can be contingent on the security of the home network. There were a few advanced users that mentioned more sophisticated network security mitigations, for example, segmenting their home network, installing virtual private networks (VPNs), or monitoring network traffic. For example, a do-it-yourselfer who customizes his smart home devices was diligent in securing his home network: *“I have a protective network where all these devices live in, and you can’t get to it from the outside. I can get to it from within my house, and if I have to I can get to it via a VPN from the outside”* (P16_A). Another also made use of VPNs *“to mask the IP address. It’s not that I’m doing anything illegal... It’s just I don’t feel like being tracked”* (P20_A).

However, most participants’ extent of network security configuration was to password-protect their Wi-Fi. One participant commented, *“When it comes to my internet that I use to connect a lot of them, you know, it is password protected. So you know, it’s not like anyone can just log on and use my network”* (P32_A). Another said, *“I’m always switching passwords with my Wi-Fi”* (P34_A).

Option Configuration. Twelve participants configured options that were at least loosely related to privacy and security. This mostly entailed disabling default functionality. For example, one participant disabled online ordering on her virtual assistant: *“We have cut off some functionality just to prevent the \$400 order of mystery items”* (P1_A). A tech-savvy participant mitigated his concerns by *“turning off certain features that I think might share more information or provide more access to the device than is necessary”* (P15_A), giving the example of how he had disabled the microphone in his smart TV. Another participant was one of the few who knew about options in virtual assistants to limit audio recording usage: *“For the [virtual assistant], it records everything. But I did see one of the options was to regularly delete it every day or something, so that kind of took the concern off the table”* (P27_A).

Limiting Shared Information. Eight participants mentioned limiting the information they share with device manufacturers, mostly when setting up companion apps. A participant said, *“I have my email address that I use for signing up for accounts that I’m never going to check and email address that I use for signing up for things that I actually care about. The latter is a very small number”* (P17_A). Another remarked, *“When it comes to, especially I think my [virtual assistant], I don’t keep certain information stored on it. Like, I know some people will keep their actual address or even sometimes even credit card information to be able to buy things right away”* (P32_A). One participant discussed

using false information when setting up her smart home device app accounts: *“I always put in fake birthdays. . . You need to know I’m eighteen, but you don’t need to know everything” (P37_A).*

Device Selection. Some participants were proactive in their mitigation efforts by considering security and privacy in their purchase decisions. One participant remarked that, prior to selecting devices, he *“paid a lot of close attention to the security of those devices and what’s happening with the data, what sorts of data they might record, how others might be able to access the system” (P15_A).* Another commented on the importance of buying secure devices: *“Even if you have to spend more money to get more into that security, we would definitely do that as we are so much dependent on this. We have to protect ourselves” (P9_A).* Others made decisions based on whether or not they trusted particular manufacturers to provide secure products. For example, a participant commented, *“I’m looking for devices that, if they’re going to communicate with a cloud service, they use a well-known cloud service” (P11_A).* One made the conscious choice to buy products from well-known, larger companies: *“These are pretty big companies. . . We’re paying money for the brand itself. . . Maybe that’s why I’m feeling a little more secure than not. . . If something happens, hopefully, they have the money to figure it out” (P6_U).*

Limiting Access. Five participants made a variety of attempts to limit access to smart home devices and their apps. Three discussed limiting access of devices by visitors and service providers entering the house. One discussed making decisions on which device to use for potentially sensitive tasks, for example, *“I don’t place orders via [my virtual assistant]. . . I do everything mostly on my computer, which has a VPN on it” (P14_U).* Another mentioned securing access to her cell-phone (which contained device companion apps) as a mitigation:

“I’m very secure with my phone. I make sure that it’s not easily accessible. . . I keep my phone right on me, I don’t set it down, I don’t let people look at stuff, I don’t access the [public Wi-Fi] internet in other areas when I’m using those apps” (P37_A).

Updates. Although updates can be a powerful mitigation against device vulnerabilities, only three participants mentioned updates or upgrades in the context of mitigations. A user said, *“I found that I’m updating everything a lot more. . . just kind of keeping up with the technology because it is so important” (P31_A).* A do-it-yourselfer purchased a smart camera with dubious ties to a foreign government, so he *“modified the firmware so it’s no longer using the [untrusted] web service or cloud service” (P11_A).*

Prior to the security and privacy portions of the interviews, we asked participants about their experiences with device updates. Participants rarely associated updates with security or privacy and mentioned that they often do not know

whether updates are available or have been installed due to inconsistent notifications and user interfaces. While updates are often viewed as potentially being security-related with traditional IT products (e.g., Microsoft’s “Patch Tuesday”), we did not find that same association in our study. In addition, users often do not apply updates if they feel their devices are still working without issue. These findings indicate both a usability problem and a perception that updates are only functionality-based and not related to security.

Lack of Mitigations. We also discovered reasons for participants not implementing mitigations. Several participants cited a lack of privacy/security options or them not being aware of available options: *“Usually the description of the controls aren’t specific enough. . . They’re like, ‘Check this for our privacy settings,’ and sometimes the description of the settings aren’t very specific”* (P13_U). Similar to reasons behind lack of concern, users often exhibited resignation and feelings of lack of control: *“I wish we could [limit data collection], but I don’t think there’ll ever be a way to control it”* (P12_U). Others cited a lack of knowledge or skill, especially with respect to cybersecurity: *“I’m not going to educate myself on network security. . . This stuff is not my forte. I’m very accepting to the fact that it is what it is”* (P8_A). Of course, some participants were simply not concerned enough to take any kind of action: *“I go on faith that they don’t find me interesting enough. I guess that’s it”* (P23_A).

5.3 Mitigation Wish List

Even though users have ultimately accepted privacy and security risks by introducing the devices into their homes, we found that they still desire greater control, especially with respect to privacy. We asked participants what they would like to do to protect their smart home privacy and security but are not doing, cannot do, or do not know how to do. Examination of the participant “wish list” provides insight into what would make users feel more empowered to take mitigating action and what options or instructional information they think manufacturers should provide.

Data Collection Transparency. Users desire manufacturers to be more forthcoming about what data is being collected, where it is going, and how it is being used (mentioned by 12 participants). Manufacturers claim that user level agreements provide this information. However, participants said that they rarely read the long agreements and generally do not find those useful because they are in *“lawyer speak. You don’t really know what they’re collecting because they can use language to mislead you”* (P31_U). The lack of transparency leaves users wanting more: *“At least give us notice in terms of who has access to it. . . We would appreciate that and make us feel more comfortable around the security behind it”* (P21_U). One user desired a more concise, clear statement of data usage: *“if these companies provided a manifesto of what information they’re interested in or how they use information and how they’re collecting information and provide*

that - a one pager - that would be great” (P2_A). Realizing that it might not be in manufacturers’ best interest to clearly disclose data usage, P31_A saw the government as having a role since “we’ve got to do something to protect people’s information, or at least make them more aware of what exactly is being utilized and sold.”

Privacy and Security Controls. Ten participants would like more control over the devices and data. This includes the ability to opt in/out of various data collections, limit how data is shared, and configure security and other privacy options. For example, a participant remarked, *“there would be some of these products that I have been avoiding purchasing that I might purchase if they provided more granular control over... all aspects of the security and privacy” (P15_A).* Another participant said he would like to be able to use two-factor authentication for his devices’ companion apps: *“There would be features that would be nice to have, I guess one being a two-factor authentication. If my phone is close to my thermostat, that’s my second factor” (P10_A).* Options should also be easy-to-configure, as mentioned by one participant: *“I think the ability to control that data should be simpler than a multi-step process” (P29_A).*

Technically advanced users were more specific about what they would like to do and wanted granular controls. A computer scientist said, *“I would really be happy actually if a lot of them had APIs [application programming interfaces] that I could use to directly program their behavior and get more control over them programmatically” (P15_A).* An electrical engineer commented:

“I’d like to have the ability to potentially allow or disallow the functionality of all these devices, maybe at given times. I’d like to be able to define what are allowable communications or protocols” (P11_A).

Five participants wished that they had the ability to keep smart home data on their local network when possible instead of the common business model of data being sent to manufacturers or their cloud services. A participant said, *“If I could not have accounts and just have it on my own home network, I would prefer that” (P17_A).* P15_A commented that he wished *“some of these devices used the voice control features locally only rather than sending clips of your voice over the Internet to be analyzed.”*

Security Feature Transparency. Four participants would like to know the level of security provided by the devices. One stated, *“it would be nice to know what security features are already there because they’re not advertised or transparent at all. And maybe to have an option to get some kind of enhanced security if you wanted to” (P24_A).* Wishing to know if he needed to bolster the security of his home network to counter potentially weak smart home security, another participant said, *“I wish I knew more about what kind of encryption they use” (P3_A).*

Assistance for Users. Within the security context, four participants expressed their desire to be provided with suggestions and instructions on how to better secure their devices. A participant unfamiliar with security best practices commented, *“I think I need to be advised on good practices that I could take... And then I probably would implement them”* (P35_A). Another suggested, *“maybe the apps that I have could throw out reminders in a more frequent manner that says are you doing something like this to protect yourself?”* (P19_A). A heavy user of smart home devices said that he would like to know how best to protect his devices against vulnerabilities: *“I would like the vulnerability identified well enough so I know what it is and then some directions on how to solve it”* (P13_A).

6 Implications

The users we interviewed were diverse in their mitigation approaches to smart home devices. Some were proactive from a privacy and security perspective and knowledgeable about the technology. Others had very little understanding of the technology and implications of use. Our results suggest that users do the best they can with the skills and the options available to them.

Most of the mitigations identified in our study were simplistic (e.g., setting passwords) or not technical in nature (e.g., placement of devices). From a privacy perspective, participants expressed the desire to be able to control what happens to their data but do not know what options are available, or, in many cases, no options exist. Security concepts and implications were more difficult for participants to grasp, with many lacking the knowledge to implement effective mitigations, for example, by properly securing their home networks. Overall, we observed that many of the participants were left with a feeling of discomfort because they had privacy and security concerns but felt powerless to address those.

Based on study results, we describe possible ways in which manufacturers could empower users to make appropriate security choices through usable interfaces and where further research may be helpful. We also identify areas that could benefit from third-party evaluation and guidance.

6.1 Considerations for Usable Security and Privacy Options

Participants’ current mitigation strategies (or lack thereof) and their wish lists for privacy and security can inform what additional options manufacturers could provide and other areas where they might alleviate user burden by defaulting to strong privacy and security.

Note that since our interview study was broader than privacy and security, we had the opportunity to delve into users’ installation and administration experiences with their smart home devices. Participants revealed that they rarely change settings after initial setup. Therefore, additional research may be warranted to investigate if installation is the best time to prompt users on security and privacy options.

Secure and Private by Default: As revealed in prior usable security research, people are often reluctant to change default security settings [28,29]. Therefore, to alleviate undue burden on users, there may be settings which manufacturers could configure to be the most secure/private by default. However, more research should be conducted to understand how setting defaults to the most secure/private options may contribute to or detract from usability.

Opt In/Out: Currently, opting out of data collection and various uses may not be possible or may be burdensome. For example, P17_A said that one manufacturer required a letter be mailed requesting to limit data sharing. Based on participants desiring more control on data usage, more research is needed regarding how manufacturers could offer easy-to-configure opt in/out options.

Data Usage Transparency: Device privacy policies and user agreements are rarely read and difficult to understand, leaving users uninformed about data collection practices. Manufacturers could provide greater transparency about what data is collected, where the data goes, how long it is stored, and who it is shared with.

Data Localization: Our participants were often concerned about manufacturer profiling of their households, selling of their data, and possible data breaches of manufacturer data storage. To counter these concerns, manufacturers could provide options to localize whatever data processing can be localized instead of sending everything to the manufacturer's cloud.

Securability: In situations where security settings might be dependent on user context, there could be a focus on “securability,” which is the “ability and knowledge to enable and configure the appropriate security features” [23]. To achieve product securability, manufacturers could facilitate secure use by providing users with real-time assistance, such as configuration wizards, to help them set the level of security appropriate for their situation. For example, users might be given the option of configuring low, medium, and high levels of security based on clear criteria (e.g., network environment, context of use, risk tolerance) gleaned through a security configuration wizard. The securability concept can also be applied to privacy settings.

Granular Options for Advanced Users: We interviewed several advanced users who were well-versed in technology and security. These users wanted more control over security settings. Therefore, in addition to supporting less technical users with guided wizards and instructions, manufacturers could offer more granular security controls for those who want them. We acknowledge that striking the right balance between an abundance of granular options and a minimal set for less-technical users may be difficult. Therefore, we recommend additional research into interface solutions that may attempt to balance these considerations.

Update Transparency: Updates are especially important as they might be the only mitigations for certain kinds of smart home device vulnerabilities (e.g., those in the code). In line with the NIST Interagency Report 8267 (Draft) Security

Review of Consumer Home Internet of Things (IoT) Products [5] recommendation that users receive update notifications in a timely manner, manufacturers might either provide an option for automatic updates or push notifications to users with clear installation instructions and descriptions of the importance of applying the update.

Network Security Tips: Home networks need to be secured to protect smart home devices. However, people often lack the knowledge and motivation to take action. For example, the FBI recommends that users segment their network [13] even though few participants in our study had the technical knowledge to be able to do so. Several of our study participants said they would like manufacturers to provide step-by-step tips on home network security (e.g., setting up secure Wi-Fi, password-protecting all devices on the network) that complement the security options provided by the devices themselves.

6.2 Third-Party Opportunities

Our results suggest that users may be open to third-party organizations (e.g., government agencies, industry groups, standards organizations) playing a bigger role in suggesting guidance for manufacturers concerning the usability of smart home security and privacy features and options. For example, the guidance produced by NIST [4, 5] provides recommendations but emphasizes that these should be tailored to specific contexts of use while not placing undue burden on the user.

The wide variety of mitigations mentioned by participants may also indicate a need for more standardization of privacy and security best practices for smart home users by trusted third parties (e.g., government agencies or an IoT industry consortium). To help users understand privacy and security implications of smart home devices, we also recommend exploring the usability considerations of having an independent, third-party ratings system similar to that which has been proposed by the Canadian Internet Society [21] and the U.S. Government Departments of Commerce and Homeland Security [22]. This ratings system would help consumers to make informed decisions about which devices to bring into their homes.

7 Limitations

In addition to typical limitations of interview studies (e.g., recall, self-report, and social desirability biases), our study may be limited in generalizability. The small sample of participants, the majority of whom were well-educated individuals living in a high-income metropolitan area, may not be fully representative of the U.S. smart home user population. However, our study population appears to mirror early adopters of smart home devices, which have been characterized in prior industry surveys [8]. We also recognize that smart home users in the U.S. may have different privacy and security attitudes from users in other countries because of political or cultural factors, for example those related to privacy expectations. Finally, our study does not capture perceptions of those choosing

not to adopt smart home technologies or limited adopters (those with only one device). Non-adopters' and limited adopters' perceptions of privacy and security could shed light on additional areas needing improvement. However, even given the limitations, our exploratory study is a solid step in investigating smart home users' perceptions and practices and can inform subsequent surveys of broader populations, for example via quantitative surveys distributed in multiple countries.

8 Conclusion

We interviewed 40 smart home users to discover their security and privacy concerns and mitigation strategies. Results indicated a number of concerns, but a willingness to accept risk in exchange for perceived benefit. Concern was sometimes, but not always, accompanied by users taking mitigating actions, although most of these actions were simplistic due to limited options or lack of user technical knowledge.

Improving the security and privacy of smart home devices will be critical as adoption of these technologies increase. Efforts should be joint between consumers, manufacturers, and third-party organizations with special consideration made for designing usable interfaces that empower users to take protective actions while not overburdening them.

Disclaimer

Certain commercial companies or products are identified in this paper to foster understanding. Such identification does not imply recommendation or endorsement by the National Institute of Standards and Technology, nor does it imply that the companies or products identified are necessarily the best available for the purpose.

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